

tf.keras.layers.MaxPooling3D crashes #51936

Closed

#51975

lugalUrim opened on Sep 11, 2021

System information

- Have I written custom code (as opposed to using a stock example script provided in TensorFlow): yes
- OS Platform and Distribution (e.g., Linux Ubuntu 16.04): Linux Ubuntu 18.04
- Mobile device (e.g. iPhone 8, Pixel 2, Samsung Galaxy) if the issue happens on mobile device: n/a
- TensorFlow installed from (source or binary): binary
- TensorFlow version (use command below): 2.6.0
- Python version: 3.6.8
- Bazel version (if compiling from source): n/a
- GCC/Compiler version (if compiling from source): n/a
- CUDA/cuDNN version: n/a
- GPU model and memory: n/a

Describe the current behavior

`tf.keras.layers.MaxPooling3D` crashes when `pool_size` contains `0`, and outputs a all-inf tensor when `pool_size` contains negative values.

Describe the expected behavior

Expect a `ValueError` to be thrown if the input `pool_size` contains zero or negative values.

Standalone code to reproduce the issue

If the `pool_size` has `0`:

```
import tensorflow as tf
pool_size = [2, 2, 0]
layer = tf.keras.layers.MaxPooling3D(strides=1, pool_size=pool_size)
input_tensor = tf.random.uniform([3, 4, 10, 11, 12], dtype=tf.float32)
res = layer(input_tensor) # crash
```

Outputs:

```
Floating point exception (core dumped)
```

If the `pool_size` has negative values:

```
import tensorflow as tf
pool_size = [2, 2, -2]
layer = tf.keras.layers.MaxPooling3D(strides=1, pool_size=pool_size,)
input_tensor = tf.random.uniform([3, 4, 10, 11, 12], dtype=tf.float32)
res = layer(input_tensor)
print(res)
```

The output is a tensor with `shape = (3, 3, 9, 14, 12)` and all `inf` values.

 **lugalUrim** added `type:bug` on Sep 11, 2021

 **google-mi-butler** assigned sushreebarsa on Sep 11, 2021

 lugalUrim on Sep 12, 2021

Author ...

It turns out that `tf.keras.layers.AveragePooling3D` also crashes when `pool_size` contains `0`.

 **sushreebarsa** added `2.6.0` `comp:keras` on Sep 12, 2021

 sushreebarsa on Sep 12, 2021

Contributor ...

@lugalUrim Please post this issue on [keras-team/keras repo](https://github.com/keras-team/keras).

To know more see;

<https://discuss.tensorflow.org/t/keras-project-moved-to-new-repository-in-https-github-com-keras-team-keras/1999>

Thank you!

 **sushreebarsa** added `stat:awaiting resp...` on Sep 12, 2021

 **WingsBrokenAngel** mentioned this on Sep 12, 2021

 [Kernel_size, pool_size should be positive integers keras-team/keras#15356](https://github.com/keras-team/keras/issues/15356)

 WingsBrokenAngel on Sep 12, 2021

...

Hi, I have created a [PR](#) for Keras to solve this issue.

 **yongtang** mentioned this on Sep 13, 2021

 [Fix crash of max_pool3d when ksize is 0 or negative #51975](https://github.com/keras-team/keras/pull/51975)

yongtang on Sep 13, 2021

Member ...

While the original issue is coming from `tf.keras.layers.MaxPooling3D`, the issue is triggered when `max_pool3d` is called directly with tensorflow itself. For that PR [#51975](#) has been created to fix the issue inside tensorflow.



sushreebarsa removed **stat:awaiting resp...** on Sep 13, 2021

lugalUrim on Sep 15, 2021

Author ...

Thanks for the information [@sushreebarsa](#), I will post new issues in the keras repo, but I guess we can keep this issue in tensorflow, as mentioned by [@yongtang](#) there are related bugs in the tensorflow operator implementations as well.

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lugalUrim on Sep 15, 2021

Author ...

Thank you [@WingsBrokenAngel](#) and [@yongtang](#) for your PRs! Actually here are more crash bugs I find when `pool_size / k_size <= 0`, so we may want to fix all of these in the codebase and also cover them in the unit tests.

- `tf.keras.layers.AveragePooling1D` , `tf.keras.layers.AveragePooling2D` , `tf.keras.layers.AveragePooling3D`
- `tf.keras.layers.MaxPooling1D` , `tf.keras.layers.MaxPooling2D` , `tf.keras.layers.MaxPooling3D`
- `tf.nn.avg_pool1d` , `tf.nn.avg_pool2d` , `tf.nn.avg_pool3d`
- `tf.nn.max_pool1d` , `tf.nn.max_pool2d` , `tf.nn.max_pool3d`

copybara-service closed this as completed in [#51975](#) on Oct 6, 2021

google-ml-butler on Oct 6, 2021

Are you satisfied with the resolution of your issue?

[Yes](#)

[No](#)



rasabot mentioned this on Nov 13, 2021

[Security Alert RasaHQ/rasa#8638](#)

oawxkw on Sep 12, 2023 · edited by oawxkw

Edits ...

I also observed the following API aliases or similar APIs can cause the same issue in some older versions of tensorflow.

Users should be cautious when using them on CPU up to tensorflow 2.6.0 (v2.6.0-rc2-32-g919f693420e).

Besides, in tensorflow 2.6.0 (v2.6.0-rc2-32-g919f693420e) and previous versions, it throws the `InternalError` Exception on GPU, which is acceptable.

- `(tf.keras.layers.MaxPooling3D)` , `tf.keras.layers.MaxPool3D` ,
`tf.compat.v1.keras.layers.MaxPooling3D` , `tf.compat.v1.keras.layers.MaxPool3D`
- `tf.keras.layers.AveragePooling3D` , `tf.keras.layers.AvgPool3D` ,
`tf.compat.v1.keras.layers.AveragePooling3D` , `tf.compat.v1.keras.layers.AvgPool3D`

► Code to reproduce the issue for the above APIs in older versions

It seems to be fixed in tensorflow 2.6.1 (v2.6.0-101-g3aa40c3ce9d) and later versions.

Note that, in tensorflow 2.6.1 (v2.6.0-101-g3aa40c3ce9d), it throws the `InvalidArgumentError` Exception, and in later versions, it throws the `ValueError` Exception, which is expected.

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Assignees



sushreebarsa

Labels

2.6.0

comp:keras

type:bug

Type

No type

Projects

No projects

Milestone

No milestone

Relationships

None yet

Development



Code with agent mode



Fix crash of max_pool3d when ksize is 0 or negative

tensorflow/tensorflow

Participants



